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The Editorial Board
Journal of Cosmology and Astroparticle Physics (JCAP)

Dear Editors,

I am pleased to submit the manuscript *"Structural Self-Energy Expansion: A Minimal-Parameter Dark-Energy Framework Anchored in φ and π"* for consideration in JCAP.

The paper addresses a central question in late-time cosmology: how much of the dark-energy phenomenology can be fixed by structure rather than by free parameters. We show that the *shape* of cosmic acceleration—the equation-of-state pair (w_0, w_a), the canonical effective-field-theory coefficients, and the inflationary spectral observables—follows algebraically from two dimensionless constants, the golden ratio φ and π, with no tuned amplitude. In CMB-likelihood terms the framework is a $k = 2$ fit: it fixes four of ΛCDM's six parameters algebraically (Ω_b, Ω_c, $n_s = 1 - \varphi^{-7}$, and the global rate $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen})$, measured input and derived screening, which lands on the pure number $3(\varphi + \pi)^2$), leaving only the primordial amplitude A_s and the optical depth τ ($k = 2$ versus $k = 6$).

The principal results that I believe merit publication are:

- The bare ratios $w_0 = -T_r/M_v = -0.839950$ and $w_a = -P_{\rm sc}/I_g = -0.669975$ match the DESI DR2 $w_0 w_a$CDM posterior at the 0.24σ level (Pantheon+; 0.2–1.8σ across SN compilations), with no fitted normalization.
- A Markov-chain analysis combining DESI BAO with the global rate $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962 \pm 0.968$ as prior yields $H_0 = 67.82 \pm 0.41$ km s^{-1} Mpc^{-1} (0.34σ from $H_0^{\rm glob}$), and the full Planck 2018 `plik` likelihood (TT, TE, EE + low-ℓ + lensing) gives $\Delta{\rm BIC} = -33.83$ relative to ΛCDM ($k = 2$ versus $k = 6$), with a best-fit χ^2 statistically indistinguishable from ΛCDM—the preference is driven by parsimony, not by a superior fit.
- The CMB matter density carries no fitted parameter: it follows from the forward identity $\omega_c = {\rm KAL}_0\, \omega_b\, n_s$, giving $\Omega_{m,{\rm CMB}} = \omega_m/h^2 = 0.308881$ in agreement with Planck at 0.88σ and reproducing the acoustic-peak structure. (The 0.41σ quoted in an earlier version is the agreement of ω_c itself, not of $\Omega_{m,{\rm CMB}}$.)
- The inflationary sector predicts $n_s = 1 - \varphi^{-7} = 0.965558$ and a tensor-to-scalar ratio $r = \varphi^{-10} = 8.130619 \times 10^{-3}$, falsifiable by LiteBIRD and CMB-S4.

In keeping with the standards of the journal, the manuscript is explicit about the framework's limits. The absolute vacuum scale enters as a single measured input rather than a derived quantity; the non-linear closure of the growth amplitude is deferred to N-body work; and the mechanism behind MIRA remains an open problem rather than a claimed result. *(Updated 2026-09-07: earlier versions of this letter said S_8 "remains in tension with weak-lensing surveys, relaxed only in the conditional two-sector extension", and cited the φ-dark-matter mass coefficient. Both are withdrawn. Against the raw KiDS-1000 data with A_s free — it is a free parameter of the model — the converged MCMC*

gives $S_8 = 0.7559 \pm 0.0189$*, which is* 0.10σ *from KiDS: there is no tension to relax. The* 3.5σ *figure was an artefact of fixing* A_s *to Planck, i.e. of importing the Planck–KiDS tension. The two-sector extension and the particle were retired on 2026-08-01.* Further updated 2026-09-20: *the same test against the KiDS-Legacy release (357 raw* $\xi_\pm$ *points; Wright et al. 2025, arXiv:2503.19441, published sixteen months before this analysis — no temporal priority is claimed) finds the shear asking for* $\log(10^{10}A_s) = 3.0255 \pm 0.0396$ *against the* 3.04483 *the CMB fixes under the same algebraic background,* 0.49σ*. Fixing* A_s *to that CMB value costs nothing (*$\chi^2 = 417.97$ *with eight free parameters, all nuisance and none cosmological, against* 418.34 *with* A_s *free;* $\Delta\mathrm{BIC} = +6.24$*), so* $S_8 = 0.8273$ *becomes a prediction rather than a fit, against* 0.8265 ± 0.0176 *measured. The three* χ^2 *compared span* 1.0*: the shear does not discriminate between the models, and I make no claim that it does.)* I have deliberately separated the algebraically fixed core from these ongoing questions so that referees can assess each on its own terms.

The work is original, has not been published elsewhere, and is not under consideration by any other journal. I have no competing interests to declare. All numerical results are reproducible from the analysis scripts accompanying the manuscript.

Thank you for your time and consideration. I would be glad to respond to any questions from the editors or referees.

Sincerely,

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